Abinsha Ibnu Ayyoob

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Education

Government Engineering College - RIT — Kottayam, Kerala
Bachelor of Technology — *Aug. 2015 – Aug. 2019*

Experience

AI Researcher & ML Engineer- Research and Development — Sep 2024 – Present
IHUB Robotics Pvt. Ltd. — *Kerala, India*

- As Principal AI Engineer, spearheaded India's first psychiatric AI web platform (METROMIND AI) on Azure. Designed and deployed a RAG pipeline using Azure Cognitive Search, Azure Vector Database, and Azure AI models to ground LLM responses, achieving 99% diagnostic accuracy and reducing diagnosis time. Led end-to-end AI pipeline and CI/CD.
- Spearheaded the "Automating Pharmacy Management" project, serving as the primary technical liaison for client communication; oversaw a team of interns, architected a data pipeline leveraging large-scale web scraping to enrich a 62,000+ entry pharma product database, and directed the preparation of diverse datasets (blister packs, invoices, prescriptions) for fine-tuning VLM and integrating LLMs for metadata processing.
- Led multiple OCR projects, designing hardware setups and optimizing pipelines to achieve 80% faster data extraction with a high degree of accuracy.
- Integrated memory-efficient computer vision solutions for the Tara Gen-1 robot, deploying models for face, emotion, and PPE detection.
- Fine-tuned LLMs on proprietary datasets for enhanced offline performance on the Tara Gen-1 robot.
- Developed and deployed a custom UNet model for fundus image segmentation, achieving 0.7875 Dice Score and 96.27% accuracy for ophthalmology disorder diagnosis.
- Engineered and deployed multiple award-winning computer vision solutions and object detection models to solve critical human-centric problems.
- Led a team in preparing over 200,000 diverse invoice images for the INDRA OCR model, a solution for extracting data from Indian invoices, significantly boosting its table extraction accuracy.

Co-founder, AI/ML Engineer — Aug. 2023 – Present
Icesmith.ai — *Karnataka, India*

- Co-founded an AI solutions startup and spearheaded the technical development of its core projects.
- Leading an ongoing research project to develop a custom Vision-Language-Action (VLA) model for zero-shot object picking in robotic manipulation within NVIDIA Isaac Lab and Isaac Sim, leveraging imitation and advanced reinforcement learning.
- Optimized and fine-tuned a Stable Diffusion model for text-to-image generation using LoRA and memory-efficient training techniques, resulting in improved image quality and faster generation times.
- Designed and implemented the "Allu Chatbot" for a mattress manufacturing firm, built with FastAPI, MySQL, Ngrok, Uvicorn, Google Cloud Console, and Google Dialogflow. This enhanced customer interactions by 25% and secured additional project opportunities.
- Engineered Machine Learning models for breast cancer classification, achieving up to 97% accuracy and improving diagnostic reliability.
- Conducted Exploratory Data Analysis (EDA) on automobile auction data, delivering actionable insights that directly influenced market decisions.

Projects

Metromind AI - Psychiatric Web App | *RAG, Azure, Django, Docker, Python, CI/CD* — Feb 2025 – Present

- Spearheaded development of India's first psychiatric AI web platform on Azure, owning the full AI pipeline.
- Engineered a tone-adaptive conversational agent using RAG to generate clinically relevant diagnosis reports.
- Built a comprehensive AI observability pipeline in Azure to enhance iteration speed, performance, and cost efficiency.

LLM Fine-tuning for Tara Gen-1 | *LLM, LoRA, Quantization, PyTorch* — Feb 2025 – Present

- Fine-tuned LLM on proprietary, augmented datasets for enhanced offline performance.

- Utilized advanced techniques including LoRA, 4-bit quantization, and 8-bit AdamW optimization for inference efficiency on the Tara Gen-1 robot.

VLA Model for Robotic Manipulation | *VLA Models, Reinforcement Learning, NVIDIA Isaac Lab/Sim* Ongoing
- Leading an ongoing research project to develop a custom Vision-Language-Action (VLA) model for zero-shot object picking.
- Developing the model within NVIDIA Isaac Lab and Isaac Sim, leveraging imitation and advanced reinforcement learning.

AI-Powered OCR Solutions | *YOLO, EasyOCR, VLM, Python, CV* Feb 2025 – Present
- Led AI-powered OCR projects for a semi-automated pharmacy setup, optimizing YOLO/EasyOCR/VLM pipelines.
- Achieved 80% faster data extraction (700+ packs/hour at <0.5% error) for medical documents and prescriptions.

Fundus Image Segmentation | *UNet, PyTorch, Computer Vision, Medical Imaging* Aug 2024 – Jan 2025
- Developed and deployed a custom UNet model for retinal vessel segmentation, achieving a 0.7875 Dice Score and 96.27% accuracy.
- Overcame dataset challenges by leveraging the DRIVE dataset with robust data augmentation and Dice BCE loss.

Stable Diffusion Fine-tuning | *LoRA, Hugging Face, Docker, Django* June 2023 – Jan 2025
- Optimized and fine-tuned a Stable Diffusion model using LoRA for text-to-image generation.
- Implemented memory-efficient training techniques and containerized the project with Docker for streamlined deployment.

Technical Skills

Languages: Python, SQL (MySQL)
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Hugging Face Transformers, Diffusers, scikit-learn, OpenCV, NLTK, spaCy, Pandas, NumPy, SciPy, Django, FastAPI
MLOps: Docker, ONNX, TensorRT, Quantization, CI/CD, Model Optimization, Ollama, Llama.cpp, GGUF, Uvicorn, Ngrok, Vector Databases
Cloud Platforms: Microsoft Azure, Google Cloud Console
Robotics & Simulation: NVIDIA Isaac Lab, NVIDIA Isaac Sim, Robotic Manipulation, Vision-Language-Action (VLA) Models, Reinforcement Learning (RL), Proximal Policy Optimization (PPO), Q-Learning, Deep Q-Learning, Imitation Learning
Key AI/ML Concepts: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Generative AI (GenAI), Diffusion Models, VAEs, Reinforcement Learning (RL), Prompt Engineering, Data Augmentation, Transfer Learning, Error Analysis, RAG, Data Visualization (Matplotlib, Seaborn)
Developer Tools: Git, VS Code, Visual Studio, PyCharm, Eclipse, Google Colab, Roboflow, LabelImg, Weights & Biases

Certificates

Machine Learning in Production Ongoing
DeepLearning.ai *Online*

Deep Learning Specialization Feb. 2024
DeepLearning.ai *Online*

Data Science and Machine Learning Mar. 2023 - June 2023
Entri Elevate *Ernakulam, Kerala*